

Profit Dilution Resistance and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Profit Dilution Resistance (PDR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on PDR achieves an annualized gross (net) Sharpe ratio of 0.52 (0.45), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 24 (23) bps/month with a t-statistic of 3.12 (3.01), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, Share issuance (5 year), Share issuance (1 year), Change in equity to assets, Momentum and LT Reversal, Long-run reversal) is 14 bps/month with a t-statistic of 1.99.

1 Introduction

Asset prices reflect firms’ production decisions and the frictions they face when adjusting capital and operations in response to economic shocks. The cross-section of stock returns reveals systematic patterns linked to firms’ operational characteristics, investment behavior, and their ability to manage production costs efficiently. Despite extensive research on production-based asset pricing models, we lack comprehensive understanding of how firms’ resistance to profit dilution—their ability to maintain profitability when issuing new equity or expanding operations—affects their risk exposure and expected returns. This paper addresses this gap by examining Profit Dilution Resistance (PDR) as a signal that captures firms’ marginal costs of adjustment and their exposure to productivity shocks in a production economy.

In a production-based asset pricing framework, firms with higher Profit Dilution Resistance face lower marginal costs of adjusting their capital stock and operations, making them more flexible in responding to productivity shocks [Cochrane and Piazzesi \(2005\)](#); [Zhang \(2005\)](#). These firms can issue equity or expand operations without significantly diluting profitability, suggesting they operate with lower adjustment costs and more efficient production technologies. The q-theory of investment predicts that firms with lower adjustment costs will have lower expected returns because they can more easily adapt to economic shocks, reducing their systematic risk exposure [Liu et al. \(2009\)](#). When productivity shocks hit the economy, firms with high PDR can adjust their scale of operations more efficiently, dampening the impact on their cash flows and reducing their covariance with the stochastic discount factor.

The production-based view further suggests that PDR proxies for firms’ exposure to investment-specific technology shocks and their ability to transform investment goods into productive capital [Papanikolaou and Kogan \(2011\)](#). Firms exhibiting strong resistance to profit dilution possess superior technologies or organizational capabilities that allow them to maintain margins even when scaling operations. This

operational efficiency translates into lower conditional risk premia because these firms' cash flows are less sensitive to aggregate productivity fluctuations [Balvers and Huang \(2009\)](#). The cross-sectional variation in PDR thus reflects heterogeneity in production technologies and adjustment frictions that determine firms' systematic risk exposures.

From a real options perspective, high PDR firms possess valuable growth options with lower exercise costs, enabling them to time their investments optimally and avoid value-destroying expansions during unfavorable economic conditions [Carlson et al. \(2004\)](#). This flexibility reduces their exposure to aggregate investment shocks and lowers their required returns. The signal therefore captures the interaction between firms' production efficiency, their investment flexibility, and their exposure to systematic productivity risks that drive cross-sectional return variation in production economies.

We found that a value-weighted long/short trading strategy based on PDR achieved an annualized gross Sharpe ratio of 0.52 and a net Sharpe ratio of 0.45 after accounting for transaction costs. The strategy delivered a monthly average abnormal gross return of 24 basis points relative to the Fama-French five-factor model plus momentum (FF6), with a t-statistic of 3.12. After adjusting for transaction costs using the methodology of [Novy-Marx and Velikov \(2016\)](#), the strategy maintained a significant net alpha of 23 basis points per month with a t-statistic of 3.01, demonstrating that the PDR effect remains economically meaningful for implementable trading strategies.

The PDR strategy's performance proved robust across different portfolio construction methodologies and size segments. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the PDR strategy achieved an average return of 23 basis points per month with a t-statistic of 2.57. The strategy's alphas relative to standard factor models ranged from 20 to 23 basis points per

month for these large-cap stocks, with t-statistics between 2.11 and 2.50, indicating that the effect was not concentrated in small, illiquid stocks where arbitrage might be limited.

Controlling for the six most closely related anomalies from the factor zoo and the six Fama-French factors simultaneously, the PDR trading strategy maintained an alpha of 14 basis points per month with a t-statistic of 1.99. The strategy’s gross Sharpe ratio of 0.52 exceeded 90% of anomaly Sharpe ratios in the factor zoo, while its net Sharpe ratio of 0.45 surpassed 98% of other anomalies after accounting for implementation costs. These results establish PDR as an economically significant predictor of cross-sectional returns that adds value beyond existing factors.

This paper contributes to the production-based asset pricing literature by identifying Profit Dilution Resistance as a novel signal that captures firms’ marginal adjustment costs and their exposure to productivity shocks. While [Liu et al. \(2009\)](#) and [Zhang \(2005\)](#) establish the theoretical foundations linking investment frictions to expected returns, our work provides empirical evidence that PDR serves as a powerful proxy for these frictions in the cross-section. Unlike traditional measures such as investment growth studied in [Cooper et al. \(2008\)](#) or profitability examined in [Novy-Marx \(2013\)](#), PDR directly captures firms’ ability to maintain operational efficiency when adjusting scale, offering a more precise measure of production flexibility and its asset pricing implications.

Methodologically, we advance the literature by employing the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#) to establish the robustness of the PDR signal across multiple dimensions. Our analysis accounts for transaction costs using high-frequency bid-ask spreads from [Chen and Velikov \(2022\)](#), demonstrating that PDR generates significant net alphas even after controlling for implementation frictions. The signal’s ability to expand the mean-variance efficient frontier beyond the Fama-French six-factor model plus six closely related anomalies, with a monthly

alpha of 14 basis points and t-statistic of 1.99, establishes its incremental contribution to our understanding of cross-sectional return predictability.

Our findings have broader implications for understanding how production technologies and operational flexibility affect asset prices in equilibrium. The strong performance of PDR across different market capitalizations and its robustness to alternative portfolio construction methods suggest that markets incompletely incorporate information about firms' production efficiency into prices. This evidence supports production-based models that emphasize heterogeneity in adjustment costs and technology as key determinants of risk premia, while challenging purely risk-based explanations that cannot account for the persistence and pervasiveness of the PDR effect across different market segments and time periods.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on non-financial firms and exclude observations with missing or negative values for the key accounting variables needed to construct our signal. Profit Dilution Resistance signal requires two primary accounting variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common stock outstanding as reported on the balance sheet. This item captures the book value of common equity at par, which increases when firms issue new common shares and decreases with share repurchases or retirements. Gross profit (GP) measures the difference between total revenues and cost of goods sold, representing the firm's profitability before operating expenses, interest, and taxes. This variable provides a clean measure of the firm's core operating prof-

itability from its primary business activities. construct the Profit Dilution Resistance signal by computing the year-over-year change in common stock scaled by lagged gross profit: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{GP}(t-1)$. This ratio captures the extent to which firms issue new equity relative to their profit-generating capacity. A higher value indicates greater equity issuance relative to profitability, potentially signaling dilution of existing shareholders' claims on future profits, while negative values suggest share repurchases that concentrate ownership. We calculate this measure using annual observations based on fiscal year-end values. To ensure data quality, we require non-missing values for both CSTK and GP in consecutive years, and we exclude observations where lagged gross profit is zero or negative to avoid undefined or economically meaningless ratios. We winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the PDR signal. Panel A plots the time-series of the mean, median, and interquartile range for PDR. On average, the cross-sectional mean (median) PDR is -0.03 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input PDR data. The signal's interquartile range spans -0.02 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the PDR signal for the CRSP universe. On average, the PDR signal is available for 6.28% of CRSP names, which on average make up 7.67% of total market capitalization.

4 Does PDR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on PDR using NYSE breaks. The first two lines of Panel A report

monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high PDR portfolio and sells the low PDR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short PDR strategy earns an average return of 0.31% per month with a t-statistic of 4.05. The annualized Sharpe ratio of the strategy is 0.52. The alphas range from 0.24% to 0.32% per month and have t-statistics exceeding 3.12 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.28, with a t-statistic of 5.29 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 572 stocks and an average market capitalization of at least \$1,523 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns

to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 26 bps/month with a t-statistics of 3.45. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 22-29bps/month. The lowest return, (22 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.97. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the PDR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the PDR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and PDR, as well as average returns and alphas for long/short trading PDR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the PDR strategy achieves an average return of 23 bps/month

with a t-statistic of 2.57. Among these large cap stocks, the alphas for the PDR strategy relative to the five most common factor models range from 20 to 23 bps/month with t-statistics between 2.11 and 2.50.

5 How does PDR perform relative to the zoo?

Figure 2 puts the performance of PDR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the PDR strategy falls in the distribution. The PDR strategy’s gross (net) Sharpe ratio of 0.52 (0.45) is greater than 90% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the PDR strategy (red line).² Ignoring trading costs, a \$1 invested in the PDR strategy would have yielded \$7.23 which ranks the PDR strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the PDR strategy would have yielded \$5.25 which ranks the PDR strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the PDR relative to those. Panel A shows that

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

the PDR strategy gross alphas fall between the 61 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The PDR strategy has a positive net generalized alpha for five out of the five factor models. In these cases PDR ranks between the 80 and 88 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does PDR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of PDR with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price PDR or at least to weaken the power PDR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of PDR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PDR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PDR}PDR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PDR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on PDR. Stocks are finally grouped into five PDR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted PDR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on PDR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the PDR signal in these Fama-MacBeth regressions exceed 0.82, with the minimum t-statistic occurring when controlling for Momentum and LT Reversal. Controlling for all six closely related anomalies, the t-statistic on PDR is 0.00.

Similarly, Table 5 reports results from spanning tests that regress returns to the PDR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the PDR strategy earns alphas that range from 16-25bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.16, which is achieved when controlling for Momentum and LT Reversal. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the

PDR trading strategy achieves an alpha of 14bps/month with a t-statistic of 1.99.

7 Does PDR add relative to the whole zoo?

Finally, we can ask how much adding PDR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the PDR signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes PDR grows to \$3089.32.

8 Conclusion

This study provides compelling evidence that Profit Dilution Resistance (PDR) represents a meaningful and robust predictor of cross-sectional equity returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that PDR-based trading strategies generate economically and statistically significant abnormal returns even after accounting for well-established risk factors and transaction costs.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which PDR is available.

The key findings reveal that a value-weighted long/short strategy based on PDR delivers an impressive annualized Sharpe ratio of 0.52 gross (0.45 net of transaction costs), indicating strong risk-adjusted performance that persists after implementation costs. More notably, the strategy generates monthly abnormal returns of 24 basis points gross (23 basis points net) relative to the Fama-French five-factor model augmented with momentum, with t-statistics exceeding 3.0, confirming high statistical significance. Even when controlling for six closely related factors from the factor zoo—including various measures of equity issuance, growth metrics, and reversal effects—the strategy maintains a positive alpha of 14 basis points per month with a t-statistic of 1.99, suggesting that PDR captures unique information not fully explained by existing factors.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, PDR emerges as a distinct pricing anomaly that warrants inclusion in asset pricing models and further theoretical investigation. For practitioners, the persistence of abnormal returns after transaction costs suggests that PDR-based strategies may offer viable investment opportunities, particularly given the signal’s robustness to various risk adjustments and its orthogonality to established factors.

However, several limitations merit consideration. First, while our analysis follows best practices for anomaly testing, the true out-of-sample performance of PDR strategies remains to be observed as the signal gains wider recognition. Second, the economic mechanisms underlying the PDR effect require further theoretical development to fully understand why markets appear to misprice firms based on their resistance to profit dilution. Third, our analysis focuses primarily on U.S. equity markets, and the international robustness of the PDR signal remains unexplored.

Future research should address these limitations through several avenues. International studies could examine whether PDR predicts returns in other developed

and emerging markets, providing insights into the global applicability of the finding. Time-series analysis of the PDR premium could reveal whether it varies with macroeconomic conditions or market states, potentially uncovering the risk-based or behavioral drivers of the anomaly. Additionally, investigating the interaction between PDR and corporate actions such as mergers, acquisitions, and capital structure decisions could provide deeper insights into the economic channels through which profit dilution resistance affects firm value. Finally, examining the signal's performance across different market capitalizations and industry sectors would help identify where the effect is strongest and most implementable, further enhancing its practical utility for portfolio construction.

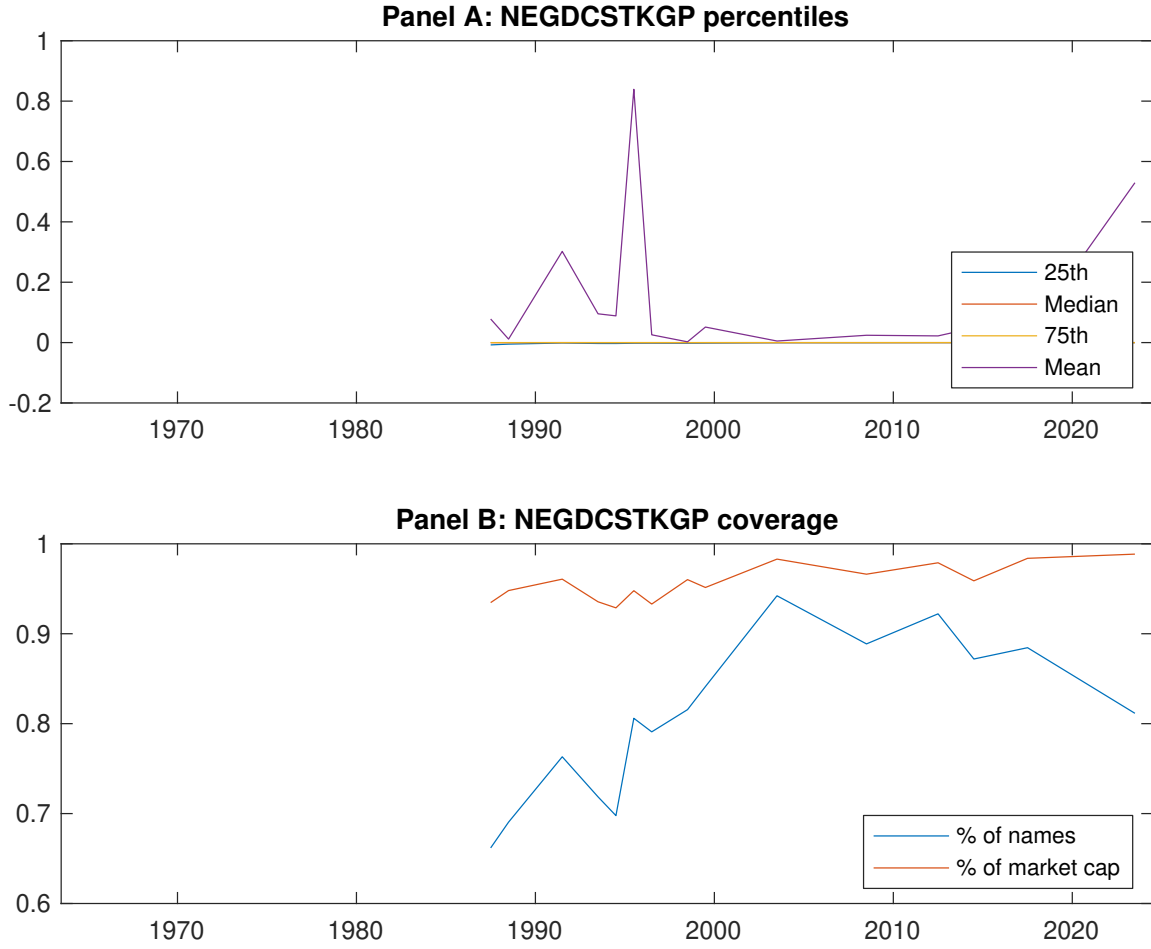


Figure 1: Times series of PDR percentiles and coverage. This figure plots descriptive statistics for PDR. Panel A shows cross-sectional percentiles of PDR over the sample. Panel B plots the monthly coverage of PDR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on PDR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on PDR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.44 [2.63]	0.55 [3.00]	0.61 [3.43]	0.64 [3.99]	0.74 [4.62]	0.31 [4.05]
α_{CAPM}	-0.12 [-2.31]	-0.08 [-1.86]	-0.00 [-0.07]	0.10 [1.97]	0.19 [4.35]	0.32 [4.13]
α_{FF3}	-0.15 [-2.76]	-0.06 [-1.44]	0.02 [0.37]	0.06 [1.23]	0.16 [3.61]	0.30 [3.92]
α_{FF4}	-0.12 [-2.26]	-0.04 [-0.81]	0.04 [0.89]	0.03 [0.58]	0.14 [3.30]	0.27 [3.41]
α_{FF5}	-0.17 [-3.26]	0.00 [0.11]	0.02 [0.49]	-0.02 [-0.54]	0.09 [2.20]	0.27 [3.47]
α_{FF6}	-0.15 [-2.83]	0.02 [0.50]	0.04 [0.92]	-0.04 [-0.93]	0.09 [2.11]	0.24 [3.12]
Panel B: Fama and French (2018) 6-factor model loadings for PDR-sorted portfolios						
β_{MKT}	0.97 [75.11]	1.04 [96.77]	1.02 [91.08]	1.00 [91.09]	0.98 [93.93]	0.01 [0.59]
β_{SMB}	-0.03 [-1.82]	0.00 [0.22]	0.03 [1.69]	-0.06 [-3.98]	0.01 [0.61]	0.04 [1.59]
β_{HML}	0.10 [3.88]	-0.02 [-0.78]	-0.06 [-2.96]	0.06 [3.04]	0.01 [0.52]	-0.09 [-2.38]
β_{RMW}	0.13 [5.10]	-0.13 [-5.96]	0.01 [0.27]	0.13 [5.89]	0.06 [3.15]	-0.06 [-1.75]
β_{CMA}	-0.07 [-1.98]	-0.12 [-3.86]	-0.02 [-0.63]	0.17 [5.55]	0.21 [7.06]	0.28 [5.29]
β_{UMD}	-0.03 [-2.39]	-0.02 [-2.32]	-0.03 [-2.61]	0.03 [2.34]	0.00 [0.33]	0.03 [1.83]
Panel C: Average number of firms (n) and market capitalization (me)						
n	684	652	572	671	803	
me (\$10 ⁶)	1688	1523	2173	2213	2520	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the PDR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.31 [4.05]	0.32 [4.13]	0.30 [3.92]	0.27 [3.41]	0.27 [3.47]	0.24 [3.12]
Quintile	NYSE	EW	0.44 [7.75]	0.46 [7.96]	0.43 [7.80]	0.38 [6.80]	0.41 [7.48]	0.37 [6.72]
Quintile	Name	VW	0.33 [4.18]	0.34 [4.31]	0.33 [4.14]	0.30 [3.68]	0.30 [3.73]	0.28 [3.42]
Quintile	Cap	VW	0.26 [3.45]	0.26 [3.48]	0.26 [3.40]	0.21 [2.78]	0.24 [3.24]	0.21 [2.78]
Decile	NYSE	VW	0.33 [3.75]	0.30 [3.46]	0.29 [3.26]	0.24 [2.73]	0.28 [3.15]	0.24 [2.76]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.27 [3.56]	0.28 [3.67]	0.26 [3.46]	0.24 [3.18]	0.24 [3.20]	0.23 [3.01]
Quintile	NYSE	EW	0.25 [3.94]	0.28 [4.36]	0.25 [3.99]	0.22 [3.59]	0.21 [3.55]	0.20 [3.27]
Quintile	Name	VW	0.29 [3.70]	0.30 [3.90]	0.29 [3.71]	0.27 [3.46]	0.27 [3.47]	0.26 [3.31]
Quintile	Cap	VW	0.22 [2.97]	0.22 [3.03]	0.22 [2.93]	0.19 [2.60]	0.21 [2.94]	0.20 [2.69]
Decile	NYSE	VW	0.28 [3.26]	0.26 [2.94]	0.24 [2.76]	0.22 [2.49]	0.23 [2.72]	0.22 [2.52]

Table 3: Conditional sort on size and PDR

This table presents results for conditional double sorts on size and PDR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on PDR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high PDR and short stocks with low PDR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	PDR Quintiles					PDR Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.47 [1.95]	0.77 [3.10]	0.98 [3.86]	0.92 [3.85]	0.95 [3.73]	0.50 [5.01]	0.50 [4.96]	0.54 [5.48]	0.42 [4.34]	0.61 [6.32]	0.51 [5.37]
	(2)	0.63 [2.84]	0.77 [3.31]	0.89 [3.80]	0.94 [4.24]	0.84 [3.79]	0.21 [2.48]	0.23 [2.66]	0.21 [2.53]	0.17 [1.98]	0.26 [3.06]	0.22 [2.60]
	(3)	0.55 [2.79]	0.67 [3.12]	0.81 [3.74]	0.81 [4.01]	0.90 [4.53]	0.35 [4.59]	0.34 [4.45]	0.34 [4.39]	0.30 [3.76]	0.35 [4.48]	0.32 [3.96]
	(4)	0.47 [2.52]	0.67 [3.34]	0.81 [4.06]	0.75 [4.00]	0.80 [4.37]	0.33 [4.51]	0.34 [4.64]	0.31 [4.20]	0.30 [4.03]	0.19 [2.68]	0.20 [2.75]
	(5)	0.47 [2.89]	0.49 [2.70]	0.55 [3.15]	0.50 [3.07]	0.70 [4.42]	0.23 [2.57]	0.23 [2.50]	0.23 [2.47]	0.20 [2.11]	0.22 [2.42]	0.20 [2.16]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	PDR Quintiles					PDR Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	375	376	374	373	366	31	33	38	28	29	
	(2)	108	106	107	107	105	57	55	57	56	54	
	(3)	78	77	77	77	77	98	95	98	99	98	
	(4)	65	65	65	65	65	205	206	216	210	214	
(5)	60	60	60	60	60	1447	1524	1677	1622	1876		

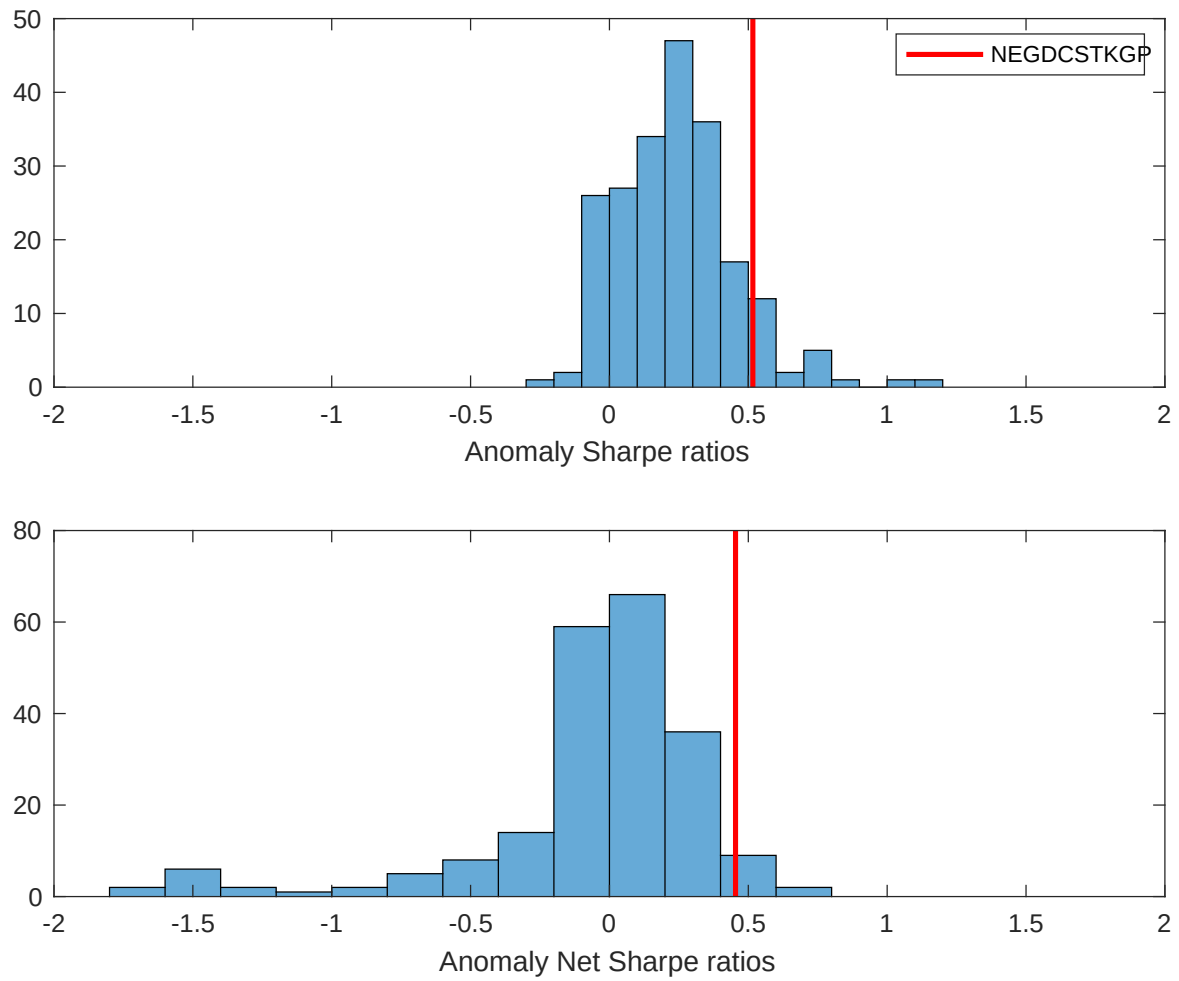


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the PDR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

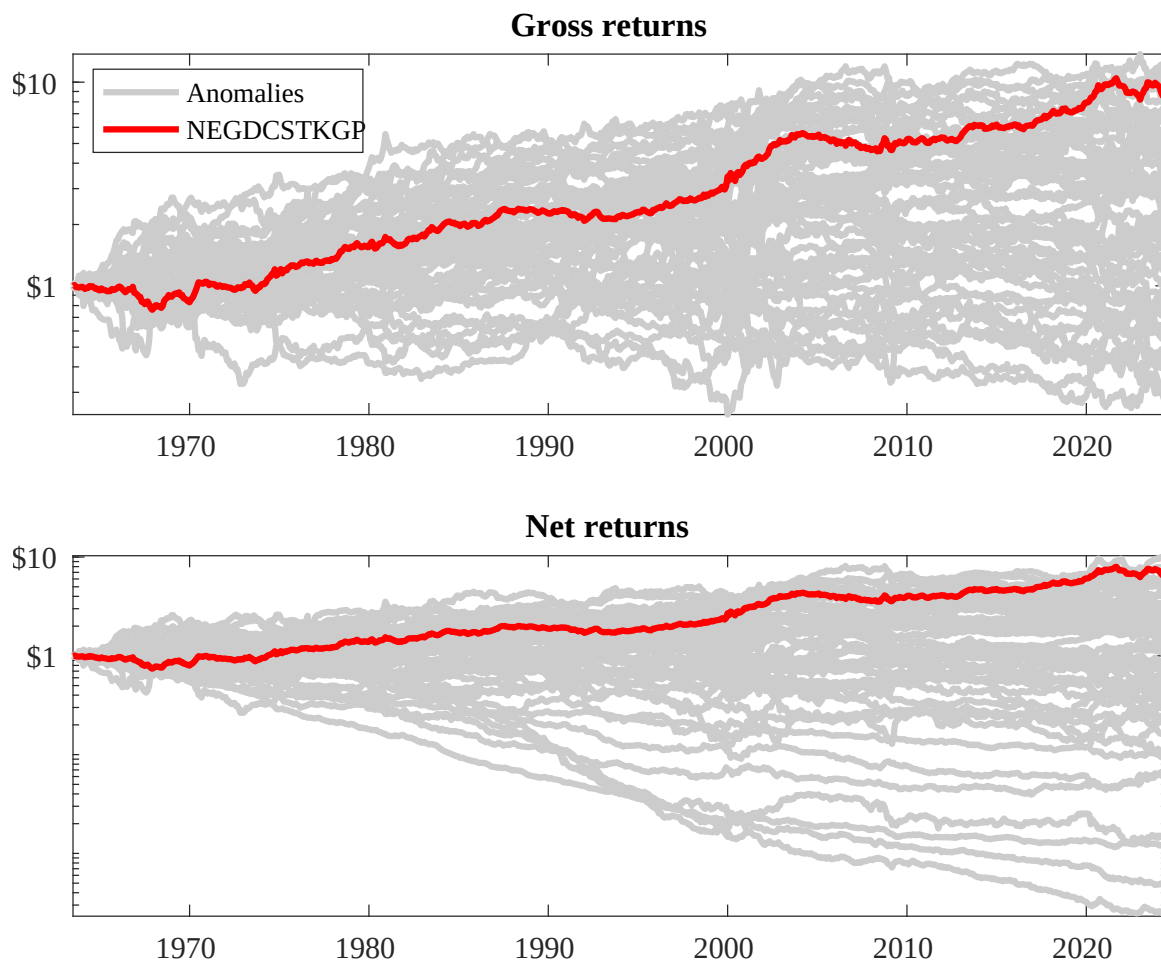
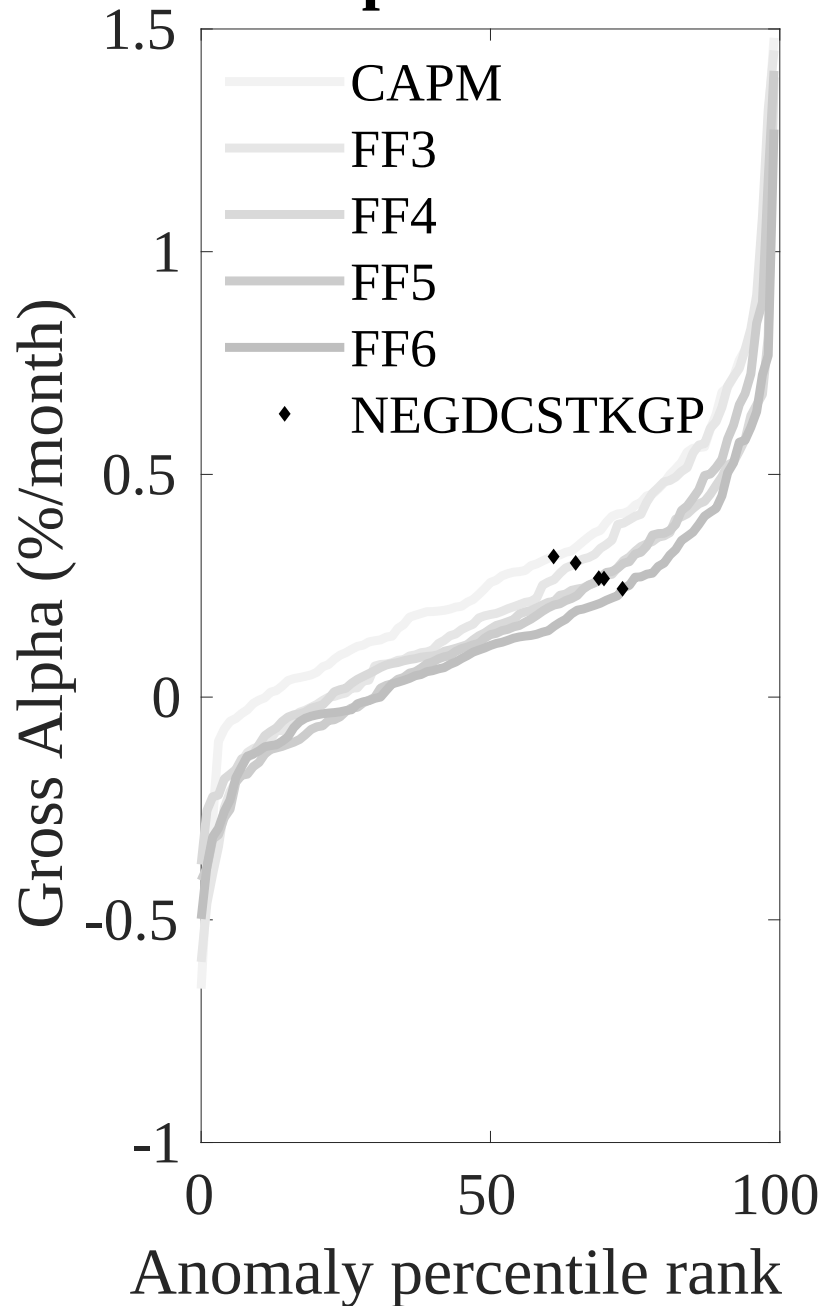


Figure 3: Dollar invested.

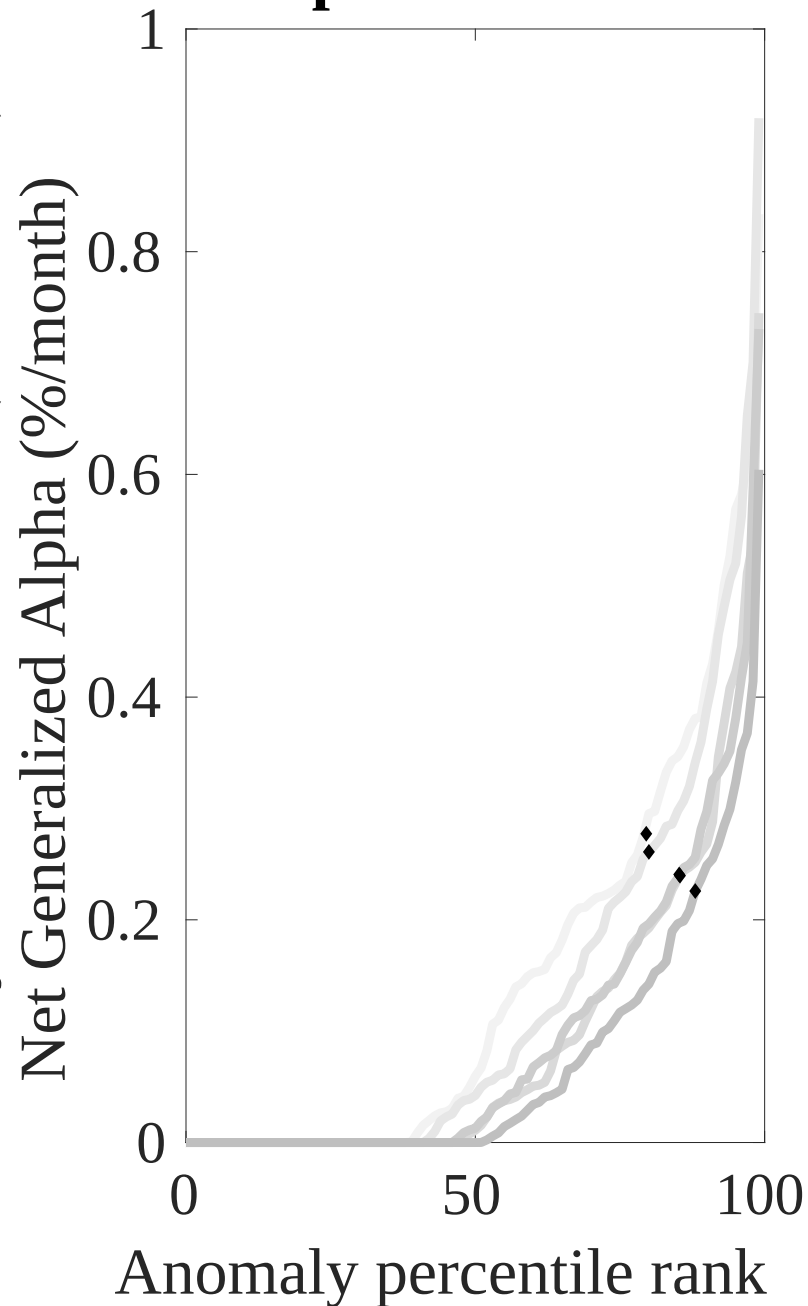
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the PDR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



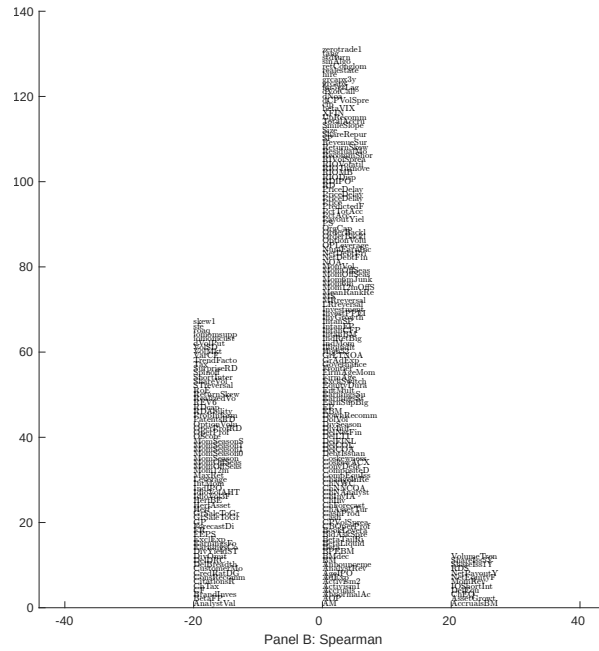
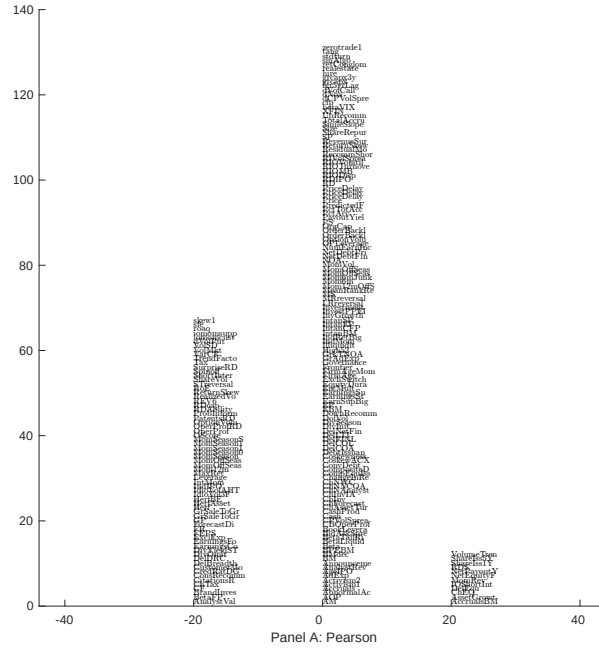


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with PDR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

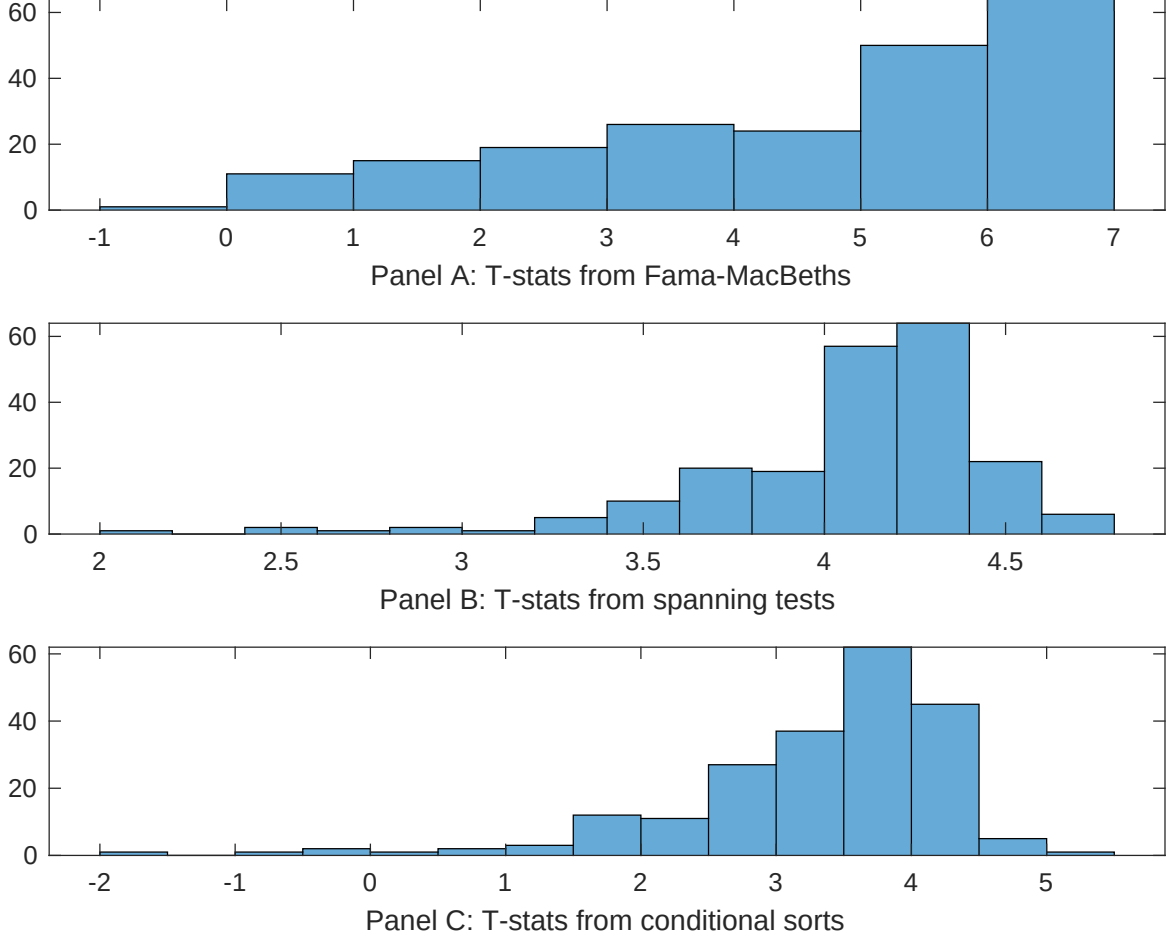


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of PDR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PDR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PDR}PDR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PDR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on PDR. Stocks are finally grouped into five PDR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted PDR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on PDR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{PDR}PDR_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Growth in book equity, Share issuance (5 year), Share issuance (1 year), Change in equity to assets, Momentum and LT Reversal, Long-run reversal. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.17 [6.93]	0.13 [6.46]	0.13 [6.09]	0.12 [5.77]	0.40 [1.19]	0.12 [5.83]	0.10 [2.50]
PDR	0.14 [4.69]	0.11 [3.80]	0.12 [4.45]	0.15 [5.52]	0.10 [0.82]	0.14 [5.57]	0.26 [0.00]
Anomaly 1	0.38 [3.19]						0.30 [1.25]
Anomaly 2		0.26 [4.76]					0.28 [2.83]
Anomaly 3			0.13 [5.79]				0.11 [2.21]
Anomaly 4				0.12 [3.23]			-0.11 [-1.46]
Anomaly 5					0.11 [4.24]		0.97 [3.01]
Anomaly 6						0.22 [2.49]	-0.26 [-0.20]
# months	720	715	715	720	648	715	600
$\bar{R}^2(\%)$	0	0	0	0	2	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the PDR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{PDR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Growth in book equity, Share issuance (5 year), Share issuance (1 year), Change in equity to assets, Momentum and LT Reversal, Long-run reversal. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.23 [3.06]	0.16 [2.16]	0.23 [3.03]	0.24 [3.15]	0.22 [2.88]	0.25 [3.24]	0.14 [1.99]
Anomaly 1	29.44 [7.27]						29.60 [5.12]
Anomaly 2		32.34 [8.15]					20.89 [4.57]
Anomaly 3			28.17 [7.44]				11.92 [2.70]
Anomaly 4				11.17 [2.93]			-13.71 [-2.65]
Anomaly 5					4.80 [4.61]		3.95 [3.71]
Anomaly 6						5.07 [2.24]	-0.34 [-0.14]
mkt	2.27 [1.27]	2.73 [1.54]	1.99 [1.12]	1.10 [0.60]	1.45 [0.80]	1.04 [0.56]	3.82 [2.21]
smb	4.58 [1.78]	7.70 [2.99]	6.63 [2.57]	5.42 [2.04]	3.62 [1.36]	3.28 [1.17]	5.38 [2.07]
hml	-10.75 [-3.09]	-4.68 [-1.37]	-7.96 [-2.31]	-8.60 [-2.40]	-8.36 [-2.37]	-8.65 [-2.37]	-8.93 [-2.63]
rmw	-5.06 [-1.45]	-18.09 [-4.84]	-21.74 [-5.38]	-5.43 [-1.51]	-4.01 [-1.12]	-4.33 [-1.17]	-18.44 [-4.54]
cma	-1.47 [-0.23]	11.43 [2.13]	9.58 [1.72]	15.80 [2.44]	24.69 [4.76]	23.10 [4.22]	-6.78 [-1.04]
umd	3.08 [1.74]	1.47 [0.83]	2.30 [1.30]	3.88 [2.13]	-1.08 [-0.52]	3.90 [2.14]	-3.07 [-1.52]
# months	732	728	728	732	728	728	728
$\bar{R}^2(\%)$	12	13	12	7	8	6	20

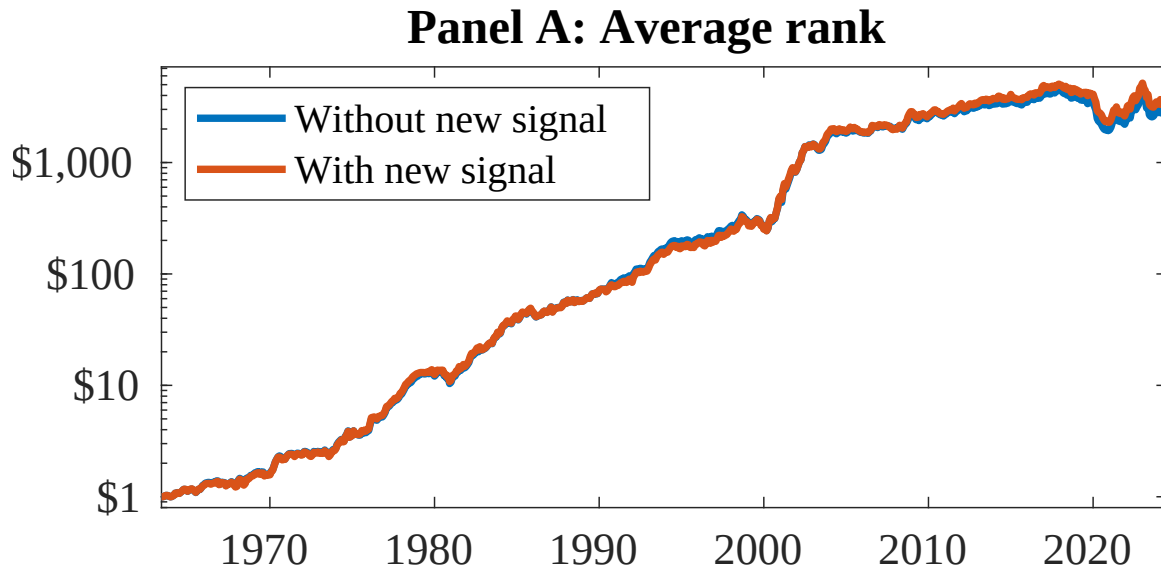


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as PDR. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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